Existing foundation assessment

25 m base-hinged high mast (Abacus type) on the base of Section K-K

Prepared by	DIMO Lighting Solutions – Design Tools (Poles & high masts, Existing foundation)
Date	20 October 2026
Status	Indicative assessment for the design team – to be checked and signed by a chartered structural engineer
Basis	BS EN 1991-1-4 wind · ILP PLG 07 hinge case · BS EN 1993-1-8 · BS EN 1992-1-1 / -4 · BS EN 1997-1

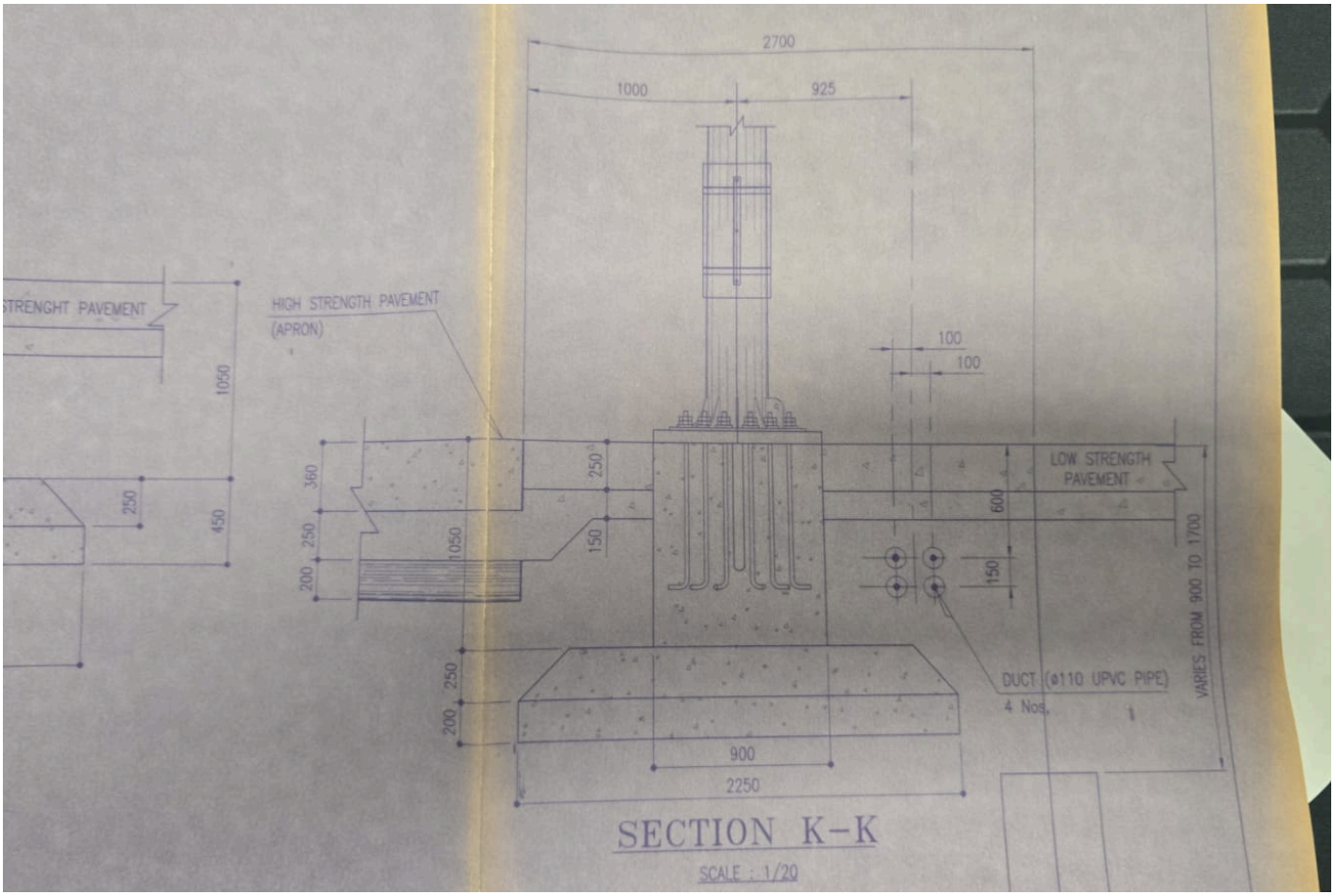
Conclusion: the existing base is NOT adequate for a 25 m base-hinged mast.

The holding-down bolts have capacity (56 % used), but the concrete block is too small and too light: overturning ratio **0.67** (needs ≥ 1.0), the soil pressure would reach about **456 kPa** at the edge with the load 1.01 m off-centre (limit B/4 = 0.56 m, allowable 150 kPa), and the pad fails one-way shear (0.53 > 0.40 MPa).

The same base **is adequate for an 18 m HL330 base-hinged mast** (as in Abacus calculation D4980C5 for BIA) – it was most likely designed for that size. A 25 m mast needs about a **2.75 × 2.75 × 0.5 m pad at 2 m with a 1.15 m pedestal (≈ 6.0 m³)**.

1. The existing base – as read from the drawing

Item	Read from Section K-K (scale 1:20, photo)	Used in the check
Pad (footing)	2250 mm wide; 200 mm at the edge, sloping up 250 mm to about 450 mm at the pedestal	2.25 × 2.25 m (assumed square), 0.45 m thick
Pedestal	900 mm square, about 1.1 m high, cast-in hooked (J) bolts	0.90 m square, top 75 mm above the apron
Depth	“Varies from 900 to 1700” below the pavement	1.6 m to the underside (deep side); 0.9 m checked as the shallow side
Holding-down bolts	Number, size and PCD not legible on the photo; embedment ≈ 850 mm	8 × M30 grade 8.8 on 620 mm PCD (Abacus GL400 pattern), hef 0.85 m
Concrete / reinforcement	Not legible	C25/30; bottom H16 @ 200 each way (assumed)
Surroundings	250 mm high-strength apron (1050 mm wide strip), low-strength pavement, 4 Ø110 uPVC ducts	Not counted (no positive tie shown)
Soil	Not given	Granular, φ′ 30°, γ 18 kN/m³, allowable bearing 150 kPa



The drawing as received (Section K-K), turned upright.

2. The mast and its loads

Item	Value
Mast	25 m base-hinged, octagonal S355 shaft 550 → 150 mm, ≈ 1050 kg; lowered by counterbalance ram
Head load	4 floodlights × 12 kg, 0.30 m² each (Cf 1.0) + head frame 30 kg, 0.15 m² (assumed) = **78 kg, 1.35 m²**
Abacus range	**GL400** at 25 m: 195 kg and 1.309 m² at 45 m/s → 1.84 m² at the site gust – carries this head; counterbalance RLH12; 8 × M30 on 620 mm PCD
Wind	Sri Lanka zone 3 (incl. Colombo) – 38 m/s 3-s gust → vb,0 26.6 m/s; 50-year return; qp at 25 m ≈ 1.03 kPa
Base actions – wind case (unfactored)	Moment **162.2 kNm**, shear 11.9 kN, vertical 11.1 kN · design moment MEd 227.1 kNm
Base actions – lowering (hinge) case	110.4 kNm unfactored (175.9 kNm with dead × 1.6) – less than the wind case, so wind governs the foundation

3. Checks of the existing base – 25 m mast

Check	Calculation	Result
Existing bolts 8 × M30 grade 8.8 on PCD 620 mm – tension EN 1993-1-8 Table 3.4	Ft,Ed 181.8 kN ≤ Ft,Rd 323.1 kN (0.56)	✓ Pass
Existing bolts – shear + tension EN 1993-1-8 6.2.2(7)	0.43	✓ Pass
Concrete cone – embedment 850 mm EN 1992-4 §7.2.1.4	one bolt 181.8 kN ≤ NRd,c 735.2 kN (cracked, γMc 1.5; spacing ≥ 3 hef assumed – else check the group)	✓ Pass
Bolt circle Fit	the new hinge base needs PCD 710 mm with 8 × M30; the existing is 620 mm with 8 × M30 – drill the new plate to the existing bolts (re-check the plate) or fit an adaptor plate	! Note
Overturning – pad 2.25 × 2.25 × 0.45 m at 1.6 m EN 1997-1 §2.4.7.2 EQU	0.9 G × B/2 ≥ 1.5 M: ratio 0.67 (G 181 kN with soil over the pad)	✗ Fail
Bearing EN 1997-1 §6.5.2	qmax 456 kPa > 150 kPa; e 1.01 m > B/4 0.56 m	✗ Fail
Sliding EN 1997-1 §6.5.3	1.5 Hk 17.9 kN ≤ 53.9 kN (tan(⅔ φ′) 0.36)	✓ Pass
Pad bending – bottom H16 @ 200 EN 1992-1-1 §6.1	MEd 144 ≤ MRd 144 kNm/m (d 367 mm, As 1005 mm²/m)	✓ Pass
Pad one-way shear EN 1992-1-1 §6.2.2	vEd 0.529 > vRd,c 0.401 MPa	✗ Fail
Condition of the existing base BS EN 12504-1 · BS EN 13791	core the concrete for its strength, expose the bolts for corrosion and thread, check level and cracks before re-use – the values above assume sound concrete of the grade entered	! Note

Overturning: 0.9 × (concrete + soil over the pad + mast weight) × B/2 ≥ 1.5 × (moment + shear × depth) – BS EN 1997-1 EQU. Bearing: pressure under the pad with the load eccentricity, at least 75 % of the pad in contact. Bolts: BS EN 1993-1-8 with the anchor-bolt shear factor; concrete cone BS EN 1992-4 (cracked concrete, γMc 1.5).

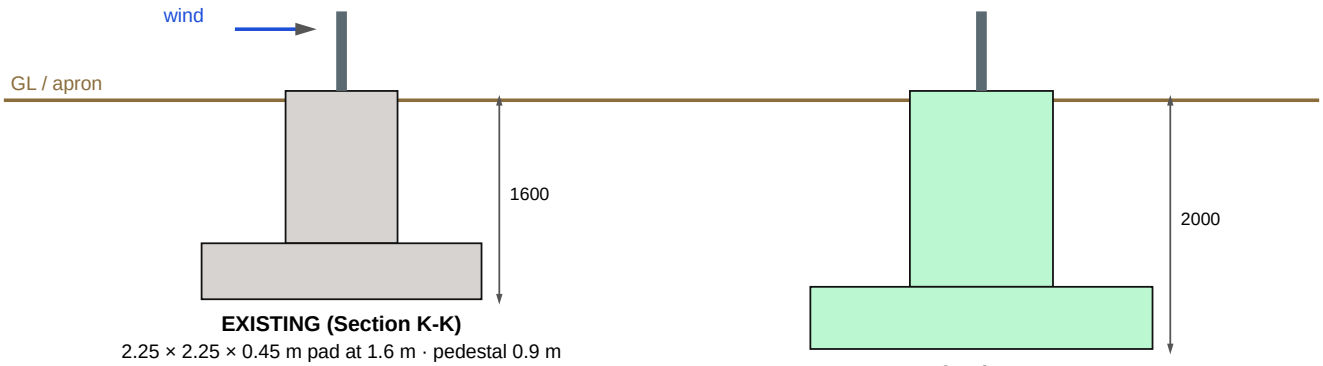
4. What the base can carry – comparison

Mast	Wind	Depth m	Base moment kNm	Overturning (≥ 1.0)	qmax kPa (≤ 150)	Bolts used	Verdict
18 m HL330 – 2 × ALO 3 + 2 × ALO 1 (as BIA calc D4980C5)	Zone 3	1.6	62.9	1.63	74	0.22	Adequate
18 m HL330 – 4 × 12 kg / 0.30 m²	Zone 3	1.6	74.6	1.38	82	0.26	Adequate
20 m – 4 × 12 kg / 0.30 m²	Zone 3	1.6	93.6	1.12	101	0.32	Not adequate – Bearing

25 m GL400 – 4 × 12 kg / 0.30 m²	Zone 3	1.6	162.2	0.67	456	0.56	**Not adequate – Overturning, Bearing, Pad one-way shear**
25 m GL400 – shallow side of the base	Zone 3	0.9	162.2	0.44	pad lifts	0.56	**Not adequate – Overturning, Bearing, Pad one-way shear**
25 m GL400 – 4 × 12 kg / 0.30 m²	Zone 2 (43 m/s)	1.6	235.4	0.47	pad lifts	0.82	**Not adequate – Overturning, Bearing, Pad one-way shear**

The Abacus calculation D4980C5 (18 m HL330, SRI LANKA BIA) gives 62 kNm at the base; this assessment gives 63 kNm for the same mast – the method agrees.

5. The base needed for the 25 m mast



EXISTING (Section K-K)
2.25 × 2.25 × 0.45 m pad at 1.6 m · pedestal 0.9 m

NEEDED for the 25 m mast
2.75 × 2.75 × 0.5 m pad at 2 m · pedestal 1.15 m

Option	What it means
A. Shorter mast on this base	An 18 m HL330 base-hinged mast passes every check on this base. 20 m fails only bearing (101 kPa with the load 0.60 m off-centre vs 0.56 m) – possible after a soil test confirms a higher bearing value.
B. Enlarge the existing base	Cast a new ring around the existing pad to reach about 2.75 × 2.75 m at 2 m depth, doweled (drilled and resin-anchored bars) into the existing pad and pedestal; the apron and ducts are cut and reinstated around it. Needs an engineer's detail for the joint.
C. New foundation	A new pad 2.75 × 2.75 × 0.5 m at 2 m with a 1.15 m pedestal and the GL400 bolt set (8 × M30 on 620 mm PCD) at each 25 m position – about 6.0 m³ of concrete.

6. To confirm before a final decision

- The pad plan shape (square or round) and the true depth at each mast position (the drawing says 900–1700 mm).
- The existing holding-down bolts: number, size, grade, PCD and condition (expose and inspect). A 460 mm PCD (HL330) will not take the GL400 base without an adaptor plate.
- Concrete strength (cores to BS EN 12504-1) and the pad reinforcement (drawing or cover-meter survey).
- Soil: allowable bearing pressure and water table from a site investigation; the 150 kPa used here is an assumption.
- The head frame weight and windage from Abacus for the chosen head (30 kg, 0.15 m² assumed), and the floodlights' tested windage at their aiming angle.
- Whether the 250 mm apron slab is tied to the pedestal – even if it is, it adds only about 50 kN, well short of the ≈ 90 kN missing for overturning.

Limitations. Indicative assessment made with DIMO Design Tools from a photograph of the drawing; dimensions not legible on the photo were assumed as stated. Abacus's own calculation for the chosen mast, head and site governs the mast and its bolts; the foundation design or strengthening is to be checked and signed by a chartered structural engineer.